

AYE
TECH HUB

• AUTOCAD • LESSON 01

Introduction to AutoCAD

The Industry-Standard 2D and 3D CAD Software

Lesson 01 of 30 | AutoCAD Program

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- What AutoCAD is and why it is used in every engineering discipline
- AutoCAD vs AutoCAD LT vs discipline-specific toolsets explained
- The DWG format: the universal language of technical drawing
- The complete AutoCAD drawing workflow from template to print
- Career pathways and salary ranges for AutoCAD-skilled engineers

What Is AutoCAD — and Why It Still Matters

AutoCAD is a **computer-aided design (CAD) software** developed by Autodesk in 1982. It remains the most widely used technical drawing software in the world, with over 1.2 billion DWG files created globally. Despite the rise of BIM tools like Revit, AutoCAD is used daily by architects, structural engineers, mechanical engineers, electrical engineers, civil engineers, and manufacturers — because it is the universal language for precise technical drawings.

Unlike BIM software which creates intelligent 3D models, AutoCAD creates **precise technical drawings** — floor plans, sections, elevations, details, and schematics — in 2D or 3D. It is faster, simpler, and more universally accepted for 2D production drawings than any alternative.

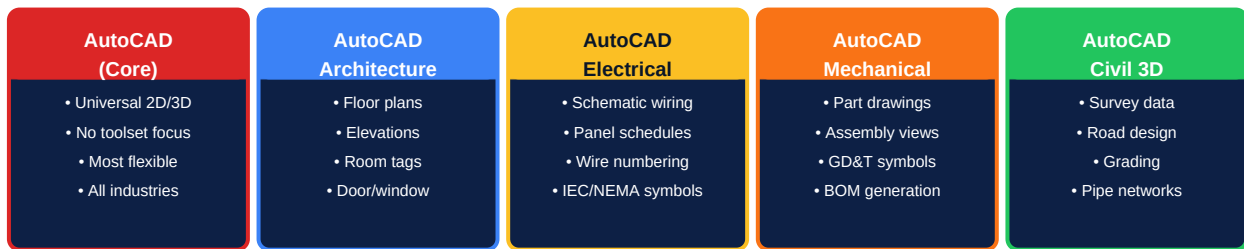


Figure 1 — The AutoCAD Product Family: Core + Discipline Toolsets

Product	Subscription Model	What It Includes	Best For
AutoCAD (full)	AutoCAD subscription ~\$250/month or \$2000/year	Full 2D/3D + all discipline toolsets (Architecture, Electrical, Mechanical, etc.)	Professionals needing multiple disciplines or switching between project types
AutoCAD LT	AutoCAD LT subscription ~\$60/month or \$510/year	2D drafting only — no 3D modelling, no scripting, no API access	Drafters focused on 2D drawing production Budget-conscious small firms
AutoCAD Architecture	Included in AutoCAD subscription	Full AutoCAD + architectural walls, doors, windows, rooms, annotation tools	Architects — faster floorplan production Section/elevation generation
AutoCAD Electrical	Included in AutoCAD subscription	Full AutoCAD + wiring diagrams, panel layouts, IEC/NEMA symbol libraries	Electrical engineers and panel designers
AutoCAD Civil 3D	Autodesk Civil 3D subscription (separate product)	AutoCAD + surfaces, alignments, profiles, corridor modelling, pipe networks	Civil/highway/drainage engineers

AutoCAD on subscription: Autodesk no longer sells perpetual licences. All AutoCAD products are subscription-based. One AutoCAD subscription now includes all discipline toolsets — Architecture, Electrical, Mechanical, Map 3D, Plant 3D, and Raster Design — at no extra cost. Students can get AutoCAD free for 1 year via the Autodesk Education Community.

The AutoCAD Workflow and DWG Format

Every professional AutoCAD project follows the same six-stage workflow. Understanding this workflow before opening the software gives you the mental map to work efficiently on any project, in any discipline.



Stage	Key Actions	AutoCAD Feature Used	Output
1 — Template Setup	Choose a .dwt template with pre-set units, layers, title block	NEW command → select template OPTIONS → Drawing Setup	Blank drawing with correct settings ready
2 — Draw	Create geometry using drawing commands: lines, arcs, polylines, circles	LINE, PLINE, ARC, CIRCLE, RECTANGLE, SPLINE	Raw geometry representing the design
3 — Modify	Edit geometry: trim, extend, copy, move, mirror, offset, array	TRIM, EXTEND, COPY, MOVE, MIRROR, OFFSET, ARRAY, FILLET	Refined, accurate geometry ready for annotation
4 — Annotate	Add dimensions, text, hatching, leaders, symbols, callouts	DIMLINEAR, MTEXT, HATCH, MLEADER, ATTDEF	Drawing communicates all information clearly
5 — Layout / Sheet	Create paper space layout, set viewport scale, arrange views	Layout tab → MVIEW → ZOOM 1/50XP (for 1:50 scale)	Sheet ready for printing at correct scale
6 — Publish / Print	Set plot style (black or colour), select printer/PDF, set scale	PLOT (Ctrl+P) or PUBLISH for batch sheets	PDF or hard copy at specified scale

The DWG Format — Why It Matters

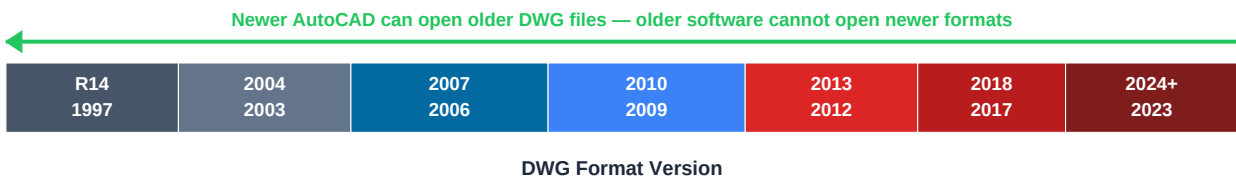


Figure 3 — DWG Format Versions: Newer Opens Older, Not Vice Versa

DWG compatibility rule: Always save in the oldest format your client or collaborator needs. If a colleague runs AutoCAD 2018, save as "AutoCAD 2013 DWG" (which opens in 2013-2023). Never save as the latest format without checking. DXF format is the universal exchange format — readable by any CAD software including free viewers and other CAD packages.

System Requirements, Careers and Getting Started

AutoCAD is more demanding than general office software but runs well on a modern mid-range laptop. The discipline-specific toolsets (Architecture, Electrical, Civil 3D) require more RAM and GPU power.

Component	Minimum (basic 2D)	Recommended (3D/large drawings)
OS	Windows 10 64-bit	Windows 11 64-bit
Processor	2.5 GHz, 4 cores	3.0 GHz+, 8 cores (rendering benefits)
RAM	8 GB	16–32 GB (Civil 3D needs 32 GB for large projects)
Storage	10 GB free (SSD preferred)	NVMe SSD — faster file operations
Graphics	1 GB VRAM, DirectX 11	4 GB+ VRAM (OpenGL 4.6 for 3D)
Display	1920×1080	2560×1440 or dual monitors (more drawing space)
Internet	Required for activation	Stable broadband for cloud storage (A360)

AutoCAD Career Pathways

Role	Core AutoCAD Skills	Salary Range (Africa, USD/yr)	Demand
CAD Drafter / Technician	2D production drawings, layers, dimensions, plotting	\$6,000–\$15,000	High — every engineering firm needs drafters
CAD Designer	Full workflow, custom blocks, customisation, collaboration	\$12,000–\$28,000	Very High — experienced designers are scarce
Structural CAD Engineer	AutoCAD + Structural knowledge; rebar drawings, shop drawings	\$15,000–\$35,000	Very High — structural drawings must be precise
Electrical CAD Engineer	AutoCAD Electrical — panel layouts, schematics, wiring	\$12,000–\$30,000	High — every electrical project needs CAD
CAD Manager / BIM Coordinator	AutoCAD + Revit + standards management + team leadership	\$25,000–\$55,000	High — firms investing in CAD/BIM need leadership

Getting Started — Your First Week	Action
Download AutoCAD free trial or student version	autodesk.com/free-software/autocad-student → 1-year free for students
Set up your workspace correctly	Use the AutoCAD Classic workspace initially → learn commands by typing, not clicking icons
Learn the 20 most essential commands	LINE, ERASE, COPY, MOVE, TRIM, EXTEND, OFFSET, FILLET, ZOOM, PAN, LAYER, DIST, ID, SNAP, ORTHO, POLAR, OTRACK, PROPERTIES, PLOT, SAVEAS
Practice drawing a simple floor plan	Draw a 5m × 4m room with 200mm walls, one door (900mm) and one window (1200mm)
Complete your first printed drawing	Set up a layout at 1:50 scale on A3 paper with a basic title block → export to PDF

What comes next — CAD-002: AutoCAD Interface — a complete tour of every panel, ribbon tab, toolbar, and command line. You will learn workspace customisation, function keys, object snap settings, and how to

configure AutoCAD for maximum productivity.